Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and computer vision: Gaussian splatting pipelines, benchmarking systems that measure every change, and continuous research that turns the best new papers into production code. Financial Engineering student at RTU.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

- 3D Gaussian Splatting reconstruction pipeline: 360° captures → navigable 3D scenes (Python, PyTorch, CUDA).
- Continuous research: read new 3DGS / computer-vision papers and implement promising methods by hand in the pipeline (e.g. fused bilateral grids, selective Adam, densification strategies), tested against the production baseline.
- Benchmarking: GPU training and evaluation runs that measure reconstruction quality for every change (ClearML, Docker, GitLab CI).
- Diffusion-based artifact fixing (Difix, then NVIDIA Fixer): repairs artifacts in rendered views of a reconstruction.
- Mesh generation from splats, including combining meshes with Gaussian splats.
- An MCP server for the team's task board, so AI agents can read and create tasks and log decisions (Node.js).
- Multi-agent AI-coding workflows for large code audits and refactors.
- Earlier (Jul–Oct 2025): a small LLM service that wrote tour narration (OpenAI API, AWS, S3).

Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

AI Talent Program (Ponder) — Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

Selective program: C++, Python, maths and AI development with industry mentors. Led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

Rīgas Valsts klasiskā ģimnāzija

Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

load-balancing-experiments (2026, Python) — seven reproducible experiments for an RTU networks project: DNS round robin and anycast measured from Riga, a queueing simulation of load-balancing policies ("power of two choices"), consistent / rendezvous / Maglev hashing, a mini load balancer with health checks, packet capture.

flow — offline workout tracker (2026, Dart/Flutter, private) — log sets with live timers; analysis of volume, load (ACWR), records and body measurements; custom formulas, JSON/CSV/PDF export, CI with golden-image tests. In daily use.

monte-carlo-finance (2026, Python) — Monte Carlo option pricing with variance reduction (up to 1,900x), GARCH volatility, VaR backtests with the Kupiec test, efficient frontier from 30,000 portfolios and CAPM; a study project from RTU courses.

Portfolio journey (this site) (2025–26, three.js) — a tap-driven WebGL world: RK4-integrated strange attractors, a single-stroke vector font, bloom, one continuous comet trail. No build step.

Offline Graphing Calculator (2023, with Ilja Volinskis) — graphing calculator that works offline; 4th place of 27 entries in the Riga Business School (RTU) programming competition for secondary school students (10 Jun 2023).

crypto-quant-lab (2026, Python) — 59 backtested versions of drift and volatility-targeting rules on 6 coins with a 1% fee. The source was recovered from Python 3.12 bytecode, and all 153 functions were verified.

projects_with_NEAT (2025, Python) — neuroevolution (NEAT) agents that learn Flappy Bird and Pong; a documented attempt at multi-snake survival.

RTU_program_parser (2025, Python/Selenium/pandas, with Aleksejs Peņkovs) — scrapes the RTU timetable and finds extra lectures that fit a student's own schedule; exports to Excel.

Snake + maze generator (2024–25, Python/Pygame) — DFS maze generation, dynamic terrain (lava, forest, rivers), BFS path-finding.

bezier-curve-js (2023, JavaScript) — interactive Bezier-curve drawing tool.

sort_visualization (2022, C++/SFML) — 16 sorting algorithms animated as columns, points, circles, spirals and pyramids, with shuffle patterns and speed control.

SKILLS

3D & vision	Gaussian splatting, 3D reconstruction, computer vision, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA
Research	Paper-to-production, benchmarking and evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, ClearML, AWS S3, Pygame, Selenium, pandas
Other	OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL
Spoken	Latvian, English, Russian